



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
PRE-SCORE
29-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 198

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **20** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

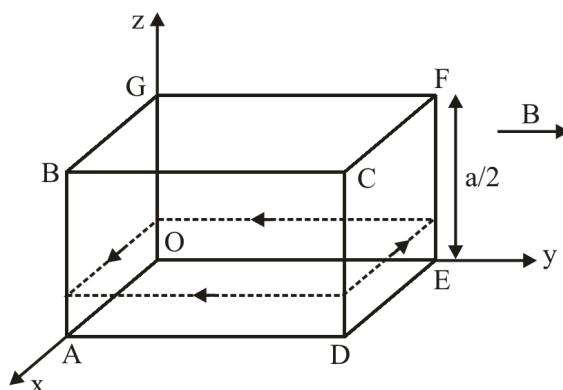
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

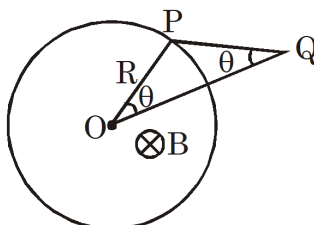
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. In an AC circuit, we have an inductance L joined in series to two resistors of resistance R each. In the first case, the resistors are in series with each other and in the second case, the resistors are parallel to each other. The power consumed by the circuit in the first case is 810W and in the second case, it is 506.25 W . If the source voltage is 225 V ,
 - (A) The value of R is $20\ \Omega$.
 - (B) In the first case, the phase difference between current through the source and voltage across it is 37° .
 - (C) In the second case, the phase difference between current through the source and voltage across it is $\tan^{-1}(3)$
 - (D) The value of Inductive reactance is $30\ \Omega$.

2. A wooden cubical block ABCDEFG of mass m and side a is wrapped by square wire loop of perimeter $4a$, carrying current I . The whole system is placed at frictionless horizontal surface in a uniform magnetic field $\vec{B} = B_0 \hat{j}$ as shown in figure. In this situation, normal force between horizontal surface and block passes through a point at a distance x from center. Choose **CORRECT** statement(s) :



- (A) The block must not topple if $I < \frac{mg}{aB_0}$
- (B) The block must not topple if $I < \frac{mg}{2aB_0}$
- (C) $x = \frac{a}{4}$ if $I = \frac{mg}{2aB_0}$
- (D) $x = \frac{a}{4}$ if $I = \frac{mg}{4aB_0}$
3. In a circular region centered at O and having radius R , magnetic field increases at a constant rate a_0 T/sec. A wire frame OPQ having resistance per unit length λ Ω/m is placed as shown in the figure. Choose the **CORRECT** option(s).



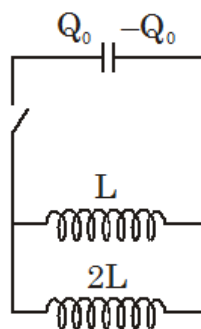
- (A) $V_P = V_Q$ for any θ as there is no induced emf across OP.

(B) $V_O - V_Q = \frac{a_0 \pi R^2}{8(\sqrt{2} + 1)}$ for $\theta = 45^\circ$

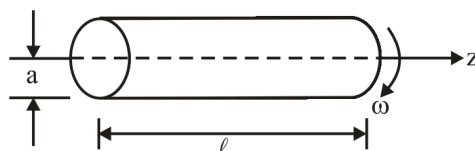
(C) $V_O - V_Q = \frac{a_0 \pi R^2}{2(\sqrt{2} - 1)}$ for $\theta = 45^\circ$

(D) $V_P - V_Q = \frac{a_0 \pi R^2}{8\sqrt{2}}$ for $\theta = 45^\circ$

4. In the figure shown, the inductors do not carry any current at the initial moment. The switch is closed at $t = 0$. There is no resistance in the circuit. Choose the correct statement (s) :



- (A) The maximum current through the inductors is at $t = \frac{\pi}{2} \sqrt{\frac{2LC}{3}}$.
- (B) The charge on left plate becomes $-Q_0/2$ at $t = \frac{5\pi}{6} \sqrt{\frac{3LC}{2}}$.
- (C) The maximum current through the inductor of inductance L is $Q_0 \sqrt{\frac{2}{3LC}}$.
- (D) The maximum rate of change of current through the inductor of inductance $2L$ is $\frac{Q_0}{2LC}$.
5. A rigid uniform stick of length L is restricted to move on the horizontal X-Y plane. A point charge q is fixed on one end of the stick and a point charge $-q$ is fixed on the other end. At a particular moment the velocity of the stick center is $\vec{v} = v_x \hat{i} + v_y \hat{j}$, the stick is at an angle θ ($< 90^\circ$) to the X-axis, and is spinning around its center at angular velocity ω in anticlockwise sense. A uniform magnetic field B is applied along the Z-axis.
- (A) The net force acting on the stick is zero.
- (B) The net force acting on the stick is $+qBL\omega (\cos \theta \hat{i} + \sin \theta \hat{j})$.
- (C) The net torque acting on the stick is zero.
- (D) The net torque acting on the stick is non-zero.
6. A cylindrical thin shell has length ℓ and radius a , where $\ell \gg a$. The surface charge density on the shell is σ . The shell rotates about its axis with an angular velocity ω which increases slowly with time as $\omega = kt$, where k is a constant.



- (A) The magnetic field inside the cylinder is $\frac{\mu_0 \sigma \omega a}{2}$.
- (B) The electric field inside the cylinder is $\frac{\mu_0 \sigma a k r}{2}$.
- (C) The magnetic energy inside the cylinder is $\frac{\pi \mu_0 \sigma^2 a^4 k^2 \ell t^2}{2}$.
- (D) The electric energy inside the cylinder is $\frac{\pi \epsilon_0 \mu_0^2 \sigma^2 k^2 a^6 \ell}{16}$.

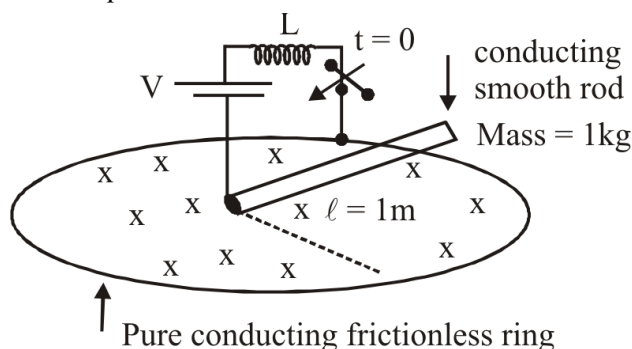
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

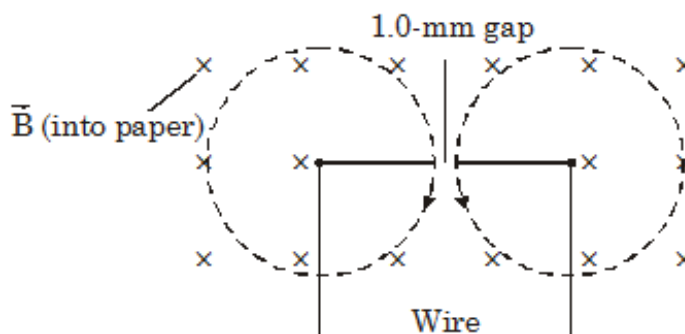
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

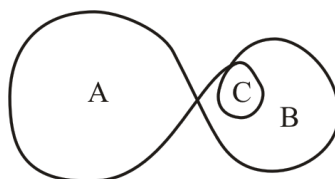
1. Consider a smooth conducting ring in horizontal plane in region of vertical uniform magnetic field, $B = 1\text{ T}$. External circuit is attached at center and periphery of conducting ring which contain battery of EMF 4 volt and inductance $L = 1\text{ H}$. Circuit is closed at $t = 0$, smooth conducting movable rod (mass 1 kg, length 1m) attaches center and periphery of ring. Neglect magnetic field produced by circuit, find the maximum angular velocity of rod (in rad/sec) in subsequent motion.



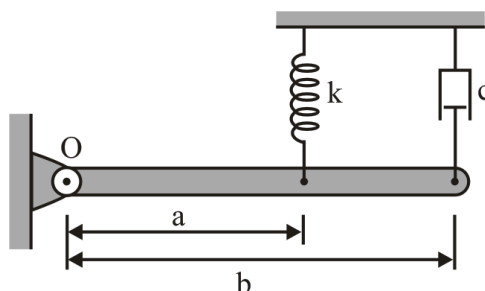
2. In an equipment, two 0.5 m long conducting rods are rotating at the same speed in opposite directions, and both are perpendicular to a 4 T magnetic field. As the drawing shows, the ends of these rods come to within 1.0 mm of each other as they rotate. Moreover, the fixed ends about which the rods are rotating are connected by a wire, so these ends are at the same electric potential. It is known that the breakdown strength of air is $3 \times 10^6\text{ V/m}$. The right rod is rotating in the clockwise direction and the left rod in the anticlockwise direction. What is the angular speed (in rad/s) of the rods when a spark jumps across the gap?



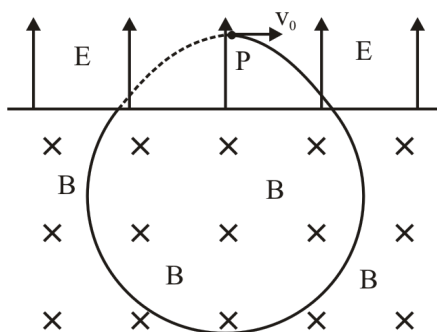
3. The following diagram depicts a single wire that is bent into the shape below. The circuit is placed in a magnetic field pointing out of the page, uniformly increasing at the rate $\frac{dB}{dt} = 2.34 \text{ T/s}$. Calculate the magnitude of induced electromotive force (in Tm^2s^{-1}) in the wire, in terms of the following labelled areas (m^2). Note that B is non-inclusive of C and that $A = 4.23$, $B = 2.74$, and $C = 0.34$.



4. A coil with 5000 turns, $r_1 = 1.25 \text{ cm}$ and $\ell_1 = 1.0 \text{ m}$ has a core with $\mu_r = 50$. A second coil of 500 turns, $r_2 = 2.0 \text{ cm}$ and $\ell_2 = 10.0 \text{ cm}$ is concentric with the first coil. The space between the coils is vacuum. The mutual inductance (in mH) is :
5. The uniform rod of mass m is freely pivoted about a horizontal axis through point O. A spring having spring constant k and a damper having damping constant c is attached to the rod as shown in the diagram. The force due to damper is equal to $-c\vec{v}$. Assuming small oscillations, the value of c (in N-s/m) for which the oscillations are critically damped is (Take : $a = 1\text{m}$, $b = 2\text{m}$, $k = 100 \text{ N/m}$, $m = 3 \text{ kg}$) (Neglect gravity)



6. A point-like negatively charged particle is projected from a point P of a uniform electric field having magnitude 100 N/C in the direction perpendicular to the electric field with a velocity v_0 . Uniform magnetic field having magnitude 10 T is also present, which is perpendicular to both the electric field vector E and to the velocity v_0 . The two types of fields are separated by a plane, which is perpendicular to the electric field vector, as shown in figure. (The whole arrangement is in vacuum, and the effect of the gravitational force on the particle is negligible). The value of v_0 for which the charge particle returns to its initial point of projection is



SECTION-III : (Maximum Marks: 18)

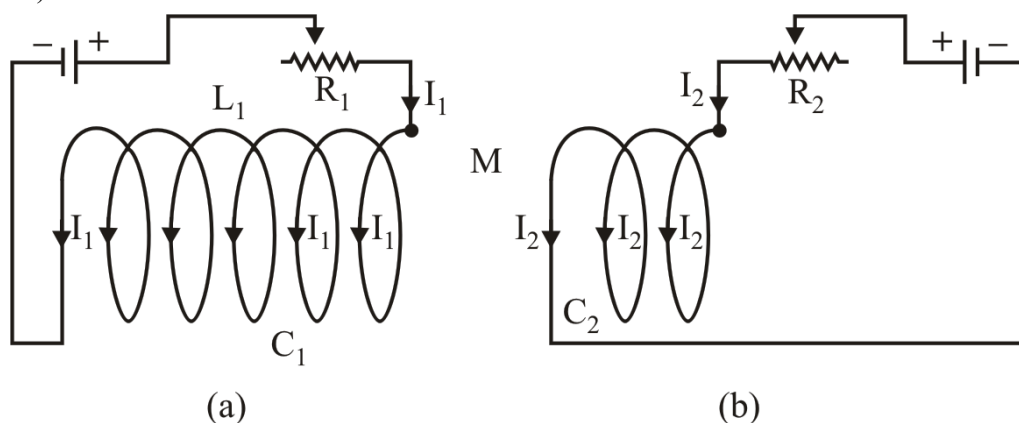
- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

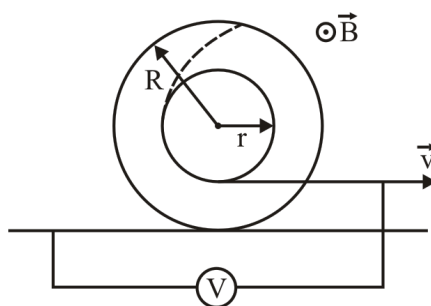
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

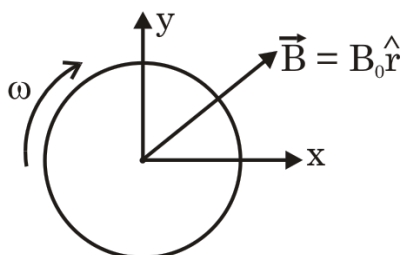
1. As seen from figure, currents I_1 and I_2 can be independently increased, decreased, or held constant by varying resistances R_1 and R_2 . Let $L_1 = 50$ mH, $L_2 = 40$ mH, $M = 15$ mH. I_1 is increased at the rate of 120 A/s and I_2 is decreased at the rate of 200 A/s. Calculate the magnitude of induced emf in coil-1 (in V).



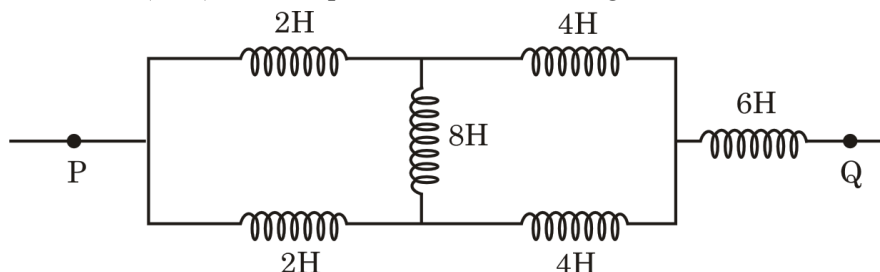
2. The coil consists of a middle cylinder with a radius r and two extreme cylinders with radii $R > r$. A long thin wire is tightly wound onto a reel as follows: first one of the outer cylinders is wound, and then the same wire is continued to be wound onto the middle cylinder in the same direction in which the winding began. After winding is completed, the coil is placed on a horizontal table placed in a uniform constant magnetic field B , whose lines of induction are parallel to the axis of the coil. An ideal voltmeter is connected to the first end of the wire lying on the table, and the other end of the wire, touching the fixed sliding contact connected to the voltmeter, is pulled along the surface of the table at a constant speed v in a direction perpendicular to the coil axis (see figure). Assuming that the coil rolls on the table without slipping, find the voltmeter reading (in V). Take : $R = 2$ m, $r = 1$ m, $v = 2$ m/s, $B = 9$ T.



3. A thin non conducting disc of mass $M = 2\text{kg}$, charge $Q = 2 \times 10^{-2}\text{C}$ and radius $R = \frac{1}{6}\text{m}$ is placed on a frictionless horizontal plane with its centre at the origin of the coordinate system. A non uniform, radial magnetic field $\vec{B} = B_0 \hat{r}$ exists in space, where $B_0 = 10\text{T}$ and $\hat{r}$ is a unit vector in the radially outward direction. The disc is set in motion with an angular velocity $\omega = x \times 10^2 \text{ rad/sec}$, about an axis passing through its centre and perpendicular to its plane, as shown in the figure. At what value of x , the disc will lift off from the surface.



4. The equivalent inductance (in H) between points P and Q in the figure is :



5. A rectangular loop ($10\text{ cm} \times 15\text{ cm}$) lies in X-Y plane with the longer side on x-axis and shorter side on y-axis and one end of the rectangle at the origin. A time dependent magnetic field $\vec{B} = B_0(x \sin \omega t \hat{j} + y \cos \omega t \hat{k})$ with $B_0 = 120 \text{ gauss/m}$ and $\omega = 1000 \text{ rad/s}$ exists within the loop. Amplitude of induced emf (in mV) is
6. A linear harmonic oscillator having a mass of 1.10 kg is set into motion with viscous damping. If the frequency is 10 Hz and if two successive amplitudes a full cycle apart are measured to be 4.65 mm and 4.30 mm , the viscous damping coefficient (in N-s/m) is (in nearest integer). For $x \ll 1$, $e^x = (1 + x)$.

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

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1. Which of the following combination of aqueous solutions of equal concentrations can result in the formation of a buffer solution ?
(A) $\text{HCl(aq)} + \text{NH}_4\text{OH(aq)}$
(B) $\text{NaOH(aq)} + \text{C}_6\text{H}_5\text{COOH(aq)}$
(C) $\text{NH}_4\text{OH(aq)} + \text{HCN(aq)}$
(D) $\text{HClO}_4\text{(aq)} + \text{KOH(aq)}$
 2. Which of the following properties of a system is/are extensive ?
(A) Standard emf of the cell
(B) pH of an acidic solution
(C) Internal energy
(D) Gibb's free energy.

3. Following data is given at 300 K :

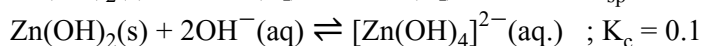
	$\Delta H_f^\circ (\text{kJ mol}^{-1})$	$S_m^\circ (\text{JK}^{-1} \text{mol}^{-1})$
$\text{CO}_2(\text{g})$	-393	214
$\text{CO}(\text{g})$	-110	200
$\text{C}(\text{graphite})$	0	6

For the reaction



- (A) $\Delta_r H^\circ = 173 \text{ kJ/mol}$ (B) $\Delta_r S^\circ = 180 \text{ Jmol}^{-1} \text{K}^{-1}$
 (C) $\Delta_r G^\circ = 53.8 \text{ kJ/mol}$ (D) $K_p^\circ < 1$

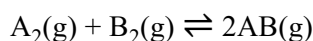
4. The salt $\text{Zn}(\text{OH})_2$ is involved in the following two equilibria,



For minimum solubility of $\text{Zn}(\text{OH})_2(\text{s})$:

- (A) $[\text{H}^+] = 10^{-10} \text{ M}$ (B) $[\text{H}^+] = 10^{-4} \text{ M}$
 (C) $[\text{OH}^-] = 10^{-4} \text{ M}$ (D) $\text{pH} = 7$

5. For the hypothetical reaction



$\Delta_r G^\circ$ and $\Delta_r S^\circ$ are 20 kJ/mol and $-20 \text{ JK}^{-1} \text{mol}^{-1}$ respectively at 200 K .

Select the correct option(s). (Use $\ln 2 = 0.7$)

[Given : $C_p(\text{AB}, \text{g}) = 40 \text{ JK}^{-1} \text{mol}^{-1}$, $C_p(\text{A}_2, \text{g}) = 25 \text{ JK}^{-1} \text{mol}^{-1}$, $C_p(\text{B}_2, \text{g}) = 35 \text{ JK}^{-1} \text{mol}^{-1}$]

- (A) ΔG° at $400 \text{ K} = 22.4 \text{ KJ mol}^{-1}$. (B) ΔH° at $400 \text{ K} = 20 \text{ KJ mol}^{-1}$.
 (C) ΔS° at $400 \text{ K} = -6 \text{ JK}^{-1} \text{mol}^{-1}$. (D) ΔH° at $200 \text{ K} = -16 \text{ KJ mol}^{-1}$.
6. pH of 0.1 M solution of NaA is 8.92 . If a drop of HPh ($\text{pK}_a = 9.52$) is added to the above solution, then :
 (It is the medical fact that our eyes can see pink colour if the molar % of ionized form of indicator is 25% or more.)
 ($\log 2 = 0.30$)
 (A) pK_a of $\text{HA} = 4.74$
 (B) Pink colour will be visible
 (C) Pink colour will not be visible
 (D) pK_a of $\text{HA} = 4.84$

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- Potassium hydrogen tartrate, $\text{KHC}_4\text{H}_4\text{O}_6$, is sometimes used in the detection of potassium. For tartaric acid, the first ionisation constant is 1.0×10^{-3} . What is the least molarity of K^+ ion which must be present in 1.0 M $\text{H}_2\text{C}_4\text{H}_4\text{O}_6$ (tartaric acid) solution for a precipitate to form? The solubility product of potassium hydrogen tartrate is 4.74×10^{-4} . (Ignore the effect of second dissociation). If the answer is $x \times 10^{-2}$ M, the value of 'x' is :
[Use $\sqrt{10} = 3.16$]
- How many milligram of sodium acetate should be added in 500 ml of 0.2 M acetic acid solution in order to make the $[\text{H}^+]$ in the solution, 4×10^{-4} M. K_a of $\text{CH}_3\text{COOH} = 1.8 \times 10^{-5}$. [Atomic mass : Na = 23]
- Calculate magnitude of work involved (in J) when 100 gm of Calcite form of CaCO_3 is converted into its Aragonite form of CaCO_3 at 100 bar pressure.
Given density of Calcite = 2gm/cc and density of Aragonite = 2.5 gm/cc.
[Atomic mass : Ca = 40]
- An amount of 0.01 moles of solid AgCN is rendered soluble in one litre of aqueous solution by adding just sufficient excess cyanide ion to form complex ion $\text{Ag}(\text{CN})_2^-$ and the concentration of free cyanide ion at equilibrium is 2.5×10^{-7} M. Determine $[\text{Ag}^+] \times 10^{10}$ in the solution neglecting hydrolysis of cyanide ion. K_{diss} for $\text{Ag}(\text{CN})_2^- = 1.0 \times 10^{-20}$.
- One mole of a monoatomic ideal gas initially at a pressure of 2.00 bar and a temperature of 273 K is taken to a final pressure of 4.00 bar by a reversible path defined by equation : $P(V)^{-3/2} = \text{constant}$, the value of $\left| \frac{\Delta U}{W} \times 100 \right|$ for this process is?
- Consider the following reversible reaction,
 $\text{A(g)} + \text{B(g)} \rightleftharpoons \text{C(g)}$
The activation energy of the backward reaction exceeds that of the forward reaction by RT (in J mol^{-1}). If the pre-exponential factor of the forward reaction is 2 times that of the reverse reaction the absolute value of ΔG^θ (in J mol^{-1}) for the reaction at 300 K is.
(Given : $\ln(2) = 0.7$, $RT = 2500 \text{ J mol}^{-1}$ at 300 K and G is the Gibbs energy)

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

- 1500 ml 0.1 M H_3PO_4 is added to 500 mL 0.1 M Na_3PO_4 . Find pH of the final solution.
Given, for H_3PO_4 : $\text{pK}_{a_1} = 2.5$, $\text{pK}_{a_2} = 7.5$ and $\text{pK}_{a_3} = 12.5$ at the given temperature.
Use : $\log 2 = 0.3$, $\log 3 = 0.5$, $\log 5 = 0.7$
- A reversible cyclic process involves 6 steps. In steps 1 and 3 system absorbs 500 J and 800 J of heat respectively from a heat reservoirs at constant temperatures of 250K and 200K respectively. Steps 2, 4 and 6, are adiabatic such that the temperature of one reservoir changes to that of next. Total work done by the system in a complete cycle is 700 J. Find the value of $\frac{T_5}{100}$ if during step 5 system exchanges heat with a reservoir at constant temperature $T_5(\text{K})$.
- An ideal gas is taken in a thermally insulated container and expanded against vacuum. Calculate the number of parameters/quantities out of the following which will be zero for this process :
 ΔU , ΔH , q , W , ΔS_{system} , $\Delta S_{\text{surrounding}}$, ΔV , ΔP , ΔG , ΔT
(Assuming that gas is not showing association or dissociation)
- K_a of HCN is 5×10^{-10} at 25°C . For maintaining a constant pH of 9, the volume (in mL) of 5M KCN solution required to be added to 10mL of 2M HCN solution is :
- A sample of an ideal monoatomic gas undergoes a reversible process following equation :
 $T^3 \cdot V^x = \text{constant}$.
$$\left[\begin{array}{c} \text{State - I} \\ T_1, V_1, P_1 \end{array} \right] \rightarrow \left[\begin{array}{c} \text{State - II} \\ T_2, V_2, P_2 \end{array} \right]$$

If $T_2 = 4 \times T_1$ and entropy of gas at both states I and II are equal, then the value of 'x' is :
- Silver ion complexes with cyanide ion to form $\text{Ag}(\text{CN})_2^-$ with formation constant, $K_f = 10^{22}$. On adding 5×10^{-3} moles of silver nitrate to 1 litre of 0.01 M NaCN solution if final concentration of Ag^+ ions in solution is $[P \times 10^{-Q}]$ then find value of $[Q - P]$?

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let A(4, -2) and B(0, 1) are two points on a parabola. Normals at A & B to the parabola intersect at P(2, 2). Tangent parallel to AB to the parabola touches the parabola at Q and tangents drawn at A & B to the parabola meet at R.
 - (A) Equation of directrix of the parabola is $2x + y + 1 = 0$
 - (B) Equation of directrix of the parabola is $y = -3$
 - (C) Length of latus rectum of parabola is $\frac{8}{5}$
 - (D) Area of triangle AQB is equal to $\frac{5}{2}$
2. Let tangent at point P on ellipse cuts directrix corresponding to focus S_1 and S_2 of ellipse at D_1 and D_2 respectively. This tangent cuts auxiliary circle at F_1 and F_2 (where $F_1 D_1 < F_2 D_1$). Lines $D_1 S_1$ and $D_2 S_2$ meet at Q. If $F_1 F_2 =$ length of semi major axis then
 - (A) $D_1 Q = D_2 Q$
 - (B) $D_1 Q \neq D_2 Q$
 - (C) $\angle PD_1 S_1 = 30^\circ$
 - (D) $\angle PD_2 S_2 = 60^\circ$

3. Let e be the eccentricity of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If $f(e)$ is the eccentricity of the conjugate hyperbola of hyperbola whose eccentricity is e , then $\int_1^3 \underbrace{f(f(\dots f(e)\dots))}_{n \text{ times}} de$ is equal to
- (A) 2, if n is even (B) 4, if n is even
(C) $2\sqrt{2}$, if n is odd (D) $4\sqrt{2}$, if n is odd
4. Let $A(at_1^2, 2at_1)$, $B(at_2^2, 2at_2)$, $C(at_3^2, 2at_3)$, be three points on the parabola $y^2 = 4ax$. If the orthocentre of triangle ABC is focus S of the parabola, then
- (A) $t_1t_2 + t_3t_2 + t_3t_1 = -5$ (B) $(1 + t_1)(1 + t_2)(1 + t_3) = -4$
(C) If $t_1 = 0$ then $t_2 + t_3 = 0$ (D) $\frac{1}{t_1t_2} + \frac{1}{t_2t_3} + \frac{1}{t_3t_1} = 1$
5. Choose the correct statement(s) :
- (A) A hyperbola whose asymptotes are $x + 2y + 3 = 0$ and $3x + 4y + 5 = 0$, is passing through the point $(1, -1)$. Then the equation of its conjugate hyperbola is $3x^2 + 10xy + 8y^2 + \lambda x + 22y + \mu = 0$ then $\mu - \lambda = 6$
- (B) The minimum and maximum distance of a point $\left(\frac{9}{5}, \frac{12}{5}\right)$ from the ellipse $4(3x + 4y)^2 + 9(4x - 3y)^2 = 900$ are λ and μ respectively then $24\mu - 7\lambda = 144$
- (C) Line $x + y = 7$ touches a parabola at $P(3, 4)$. If the focus of the parabola is $(1, 1)$ and its vertex is (λ, μ) then $\lambda\mu = 12$
- (D) If the length of the latus rectum of the parabola $169\{(x - 1)^2 + (y - 3)^2\} = (5x - 12y + 19)^2$ is $\frac{\lambda}{\mu}$ (where λ and μ are relatively primes) then $\mu + \lambda = 37$
6. Consider a parabola $y^2 = \ell x$ ($\ell > 0$) with its vertex at the origin 'O' then which of the following statement(s) is(are) true ?
- (A) If origin is joined to any point P on the given parabola and a line perpendicular to OP through P meet the axis at Q , then projection of PQ on the axis is equal to ' ℓ '
- (B) A normal is drawn at any arbitrary point P on the parabola which meets the axis at G . Locus of the middle point of PG is a parabola whose latus rectum is $\frac{\ell}{4}$
- (C) A variable parabola of latus ractum ℓ , touches the given parabola, the axes of the two parabolas being parallel. The locus of the vertex of the variable parabola is another parabola whose latus rectum is ' 2ℓ '
- (D) The locus of the mid point of the focal radii of a variable point P moving on the given parabola, is a parabola whose latus rectum is ' 4ℓ '

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let an ellipse $E_1 : \frac{x^2}{64} + \frac{y^2}{48} = 1$ and two parabolas P_1 and P_2 both passing through ends of minor axis of E_1 . Vertex of each of two parabolas lie on major axis of E_1 . If $x - 2y + 16 = 0$ is common tangent of all the three curves and A be the area bounded by P_1 and P_2 then $\frac{A}{\sqrt{3}}$ is
2. An ellipse is inscribed in the circle $x^2 + y^2 = 9$ whose major axis is $3x - 4y = 0$. From Point $(0, 3)$ a perpendicular is drawn to the line $3x - 4y = 0$ intersecting the ellipse at P and line at Q such that $PQ = 1$. If d is the distance between both the foci of ellipse then $\frac{288 d^2}{119}$ is
3. If (α, β) is the centroid of a triangle PQR where P, Q and R be three points lying on the parabola $(y - 4)^2 = 15(x + 3)$ and normals at P and Q meet at R. Then the largest integral value of m satisfying $\alpha + \beta > m$ for all possible values of α and β , is
4. Let $\vec{a} = (x^2 + 3)\hat{i} - 3y^2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} - 4x\hat{k}$ be perpendicular vectors for $x, y \in \mathbb{R}$ and the relation obtained is x and y represent a conic C. Let m and n are number of integers in the range of values of x and y respectively and if P(x, y) lies on conic C then the equation of director circle of the circle which is tangent to conic C at M(3, 0) and passing through the centre of conic is $ax^2 + by^2 - cx + 23 = 0$ then the value of $(m^2 + n^2 + a + b + c)$ is

5. Consider the following statements :

Statement-I : If length of the common tangent of circle $x^2 + y^2 = 4$ and conic $3x^2 + 5y^2 = 15$ is ℓ and $A = \frac{1}{\ell}$.

Statement-II : The number of possible tangents which can be drawn to the curve $4x^2 - 9y^2 = 36$, which are perpendicular to the straight line $5x + 2y - 10 = 0$ is B.

Statement-III : With one focus of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ as the centre, a circle is drawn which is tangent to the hyperbola with no part of the circle being outside the hyperbola. The radius of the circle is C.

Statement-IV : For each positive integer n, consider the point P with abscissa n on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line $y = x$ then $\lim_{n \rightarrow \infty} (n \cdot d_n)$ has the value equal to D.

Then the value of $(A + B + C) \div D^2$ is

6. Let A and B be two points on the major axis of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, which are equidistant from the centre. If C and D are the images of these points in the line mirror $y = mx$, $m \neq 0$ then the maximum area of quadrilateral ACBD is

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. An isosceles $\triangle ABC$ is inscribed in the circle $x^2 + y^2 = 16$. From points A, B and C three ordinates are drawn which cut the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at the points P, Q and R respectively such that A and P on the opposite sides, Q and B are on the same side and C and R are on the same side of the major axis. If $\triangle PQR$ is right angled at P and θ is the smallest angle of the $\triangle ABC$, then the value of $(4\tan\theta + 3)$ is
2. If tangent drawn to the parabola $y^2 = -ax$ ($a > 0$) from a point A(1, 0) which also touches the hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$ at B such that $\angle ASB = 90^\circ$, where S is the focus of the hyperbola then the value of $(a - b^2)$ is
3. Tangents are drawn from any point on straight line $y = x + 8$ to the auxiliary circle of $x^2 + 16y^2 = 16$. If A and B are the points of contact of these tangent and P, Q are corresponding points of A and B on the ellipse respectively. If locus of mid point of PQ is $x^2 + \alpha y^2 + \beta x + \gamma y = 0$ then the value of $(\alpha - \beta + \gamma)$ is :-
4. A ray of light emerging from the point P(0, 4) along the tangent of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and strike a circle of radius $\sqrt{65}$ units and reflected from it. The reflected ray entering inside the ellipse through the focus $(-3, 0)$ and strike lower surface of ellipse and gets reflected from it. If the reflected ray returns to the point P(0, 4), then ordinate of the centre of this circle is
5. From a point common tangents are drawn to the circle $x^2 + y^2 = \frac{a^2}{2}$ and parabola $y^2 = 4ax$. If the area of the quadrilateral formed by the common tangents, the chord of contact of the circle and the chord of contact of the parabola is $\frac{Aa^2}{B}$ (where A and B are coprime) then the value of $(A - 2B)$ is
6. A triangle is circumscribed to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and two of the vertices lie on the directrices and if the locus of the third vertex is the ellipse $b^2y^2 = a^2(e^2 + \lambda)(a^2 - \mu x^2)$ then the value of $(\lambda + \mu)$ is

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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